



Build a Virtual Private Cloud (VPC)

IR

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NextWork - Build a Virtual Private Cloud (VPC) | eu-north-1

eu-north-1.console.aws.amazon.com/vpcconsole/home

A free online introduction to VPC

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VPC > Your VPCs > Create VPC

Resources to create Info
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional Info
Creates a tag with a key of 'Name' and a value that you specify.

Rahim-NextWor-VPC

IPv4 CIDR block Info
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/24
CIDR block size must be between /16 and /28.

IPv6 CIDR block Info
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy Info
Default

VPC encryption control (\$) Info
Monitor mode provides visibility into encryption status without blocking traffic. Enforce mode prevents unencrypted traffic. [Additional charges apply](#)

☒ None ☐ Monitor mode ☐ Enforce mode
See which resources in your VPC are unencrypted but allow the creation of Requires all resources, except exclusions, in your VPC to be encryption-capable and

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Introducing Today's Project!

I'm doing this project to learn the basics of setting up a VPC network and get the insights on it to improve my knowledge of this technology.

What is Amazon VPC?

Amazon VPC is a service in Amazon which lets you create private, isolated networks on the cloud where you can run your EC2 instances. It is useful because it gives you control over IP ranges, subnets, and security so that you can securely manage your cloud infrastructure.

In today's project, I used Amazon VPC to create subnets and attach it to my VPC. Also, I learnt on how to connect my VPC to internet using Internet Gateway.

Personal reflection

This project took me around 35 minutes to complete including the secret mission.

One thing I didn't expect in this project was how everything was explained in a very easy and clear manner.

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Virtual Private Clouds (VPCs)

What I did in this step

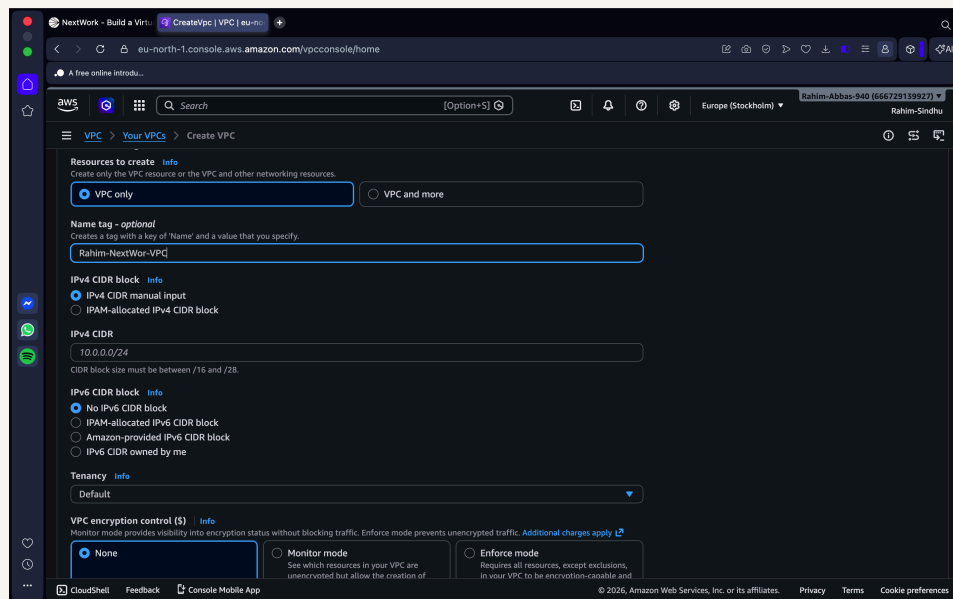
In this step, I will learn on how to create a custom VPC, learning about IPv4 addresses and CIDR.

How VPCs work

VPCs are a resource that makes sure that everything on the cloud is private to every user. Without VPCs, every file uploaded to the cloud would be available to everyone without any privacy.

Why there is a default VPC in AWS accounts

The availability of a default VPC when you create a new AWS account is to allow the user to utilize the other services like AWS Lambda, S3 bucket etc from day 1 without any hassle. If the default VPC was not available, the user would have to learn on how to create a VPC and that would take them some time provided if they didn't have prior knowledge of it.



Defining IPv4 CIDR blocks

To set up my VPC, I had to define an IPv4 CIDR block, which is a collection/network of IPv4 addresses. IPv4 CIDR have a collection of unique IP addresses belongs to a network and is used for routing and organization.



Subnets

What I did in this step

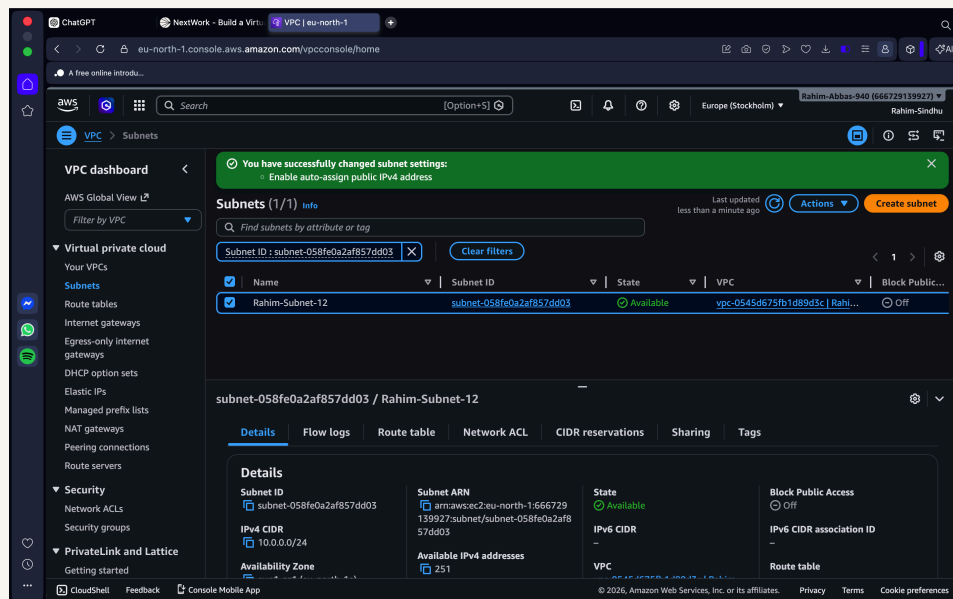
In this step, I will be learning on how to create subnets in my custom VPC

Creating and configuring subnets

Subnets are smaller ranges of IP addresses within a VPC. They help organize resources and control network access. A subnet can be public or private depending on its routing configuration. Each subnet must use a unique, non-overlapping CIDR block within the VPC.

Public vs private subnets

The difference between public and private subnets are that or a subnet to be considered public, it has to be directly accessible from the internet. A private subnet can only be accessed within the organization or the EC2 instance in which it is created. A public subnet can directly communicate with the internet using an internet gateway while a private subnet does not have direct access to the internet and is typically used for internal access outside of the internet's reach



Auto-assigning public IPv4 addresses

The need to enable this setting is that by default, the subnet is set to private meaning that it can not be accessed from the internet. To allow the access to the subnet via the internet, we have to enable the "auto-assign public IPv4 addresses" that would make the privacy setting of the subnet from private to public.

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Internet gateways

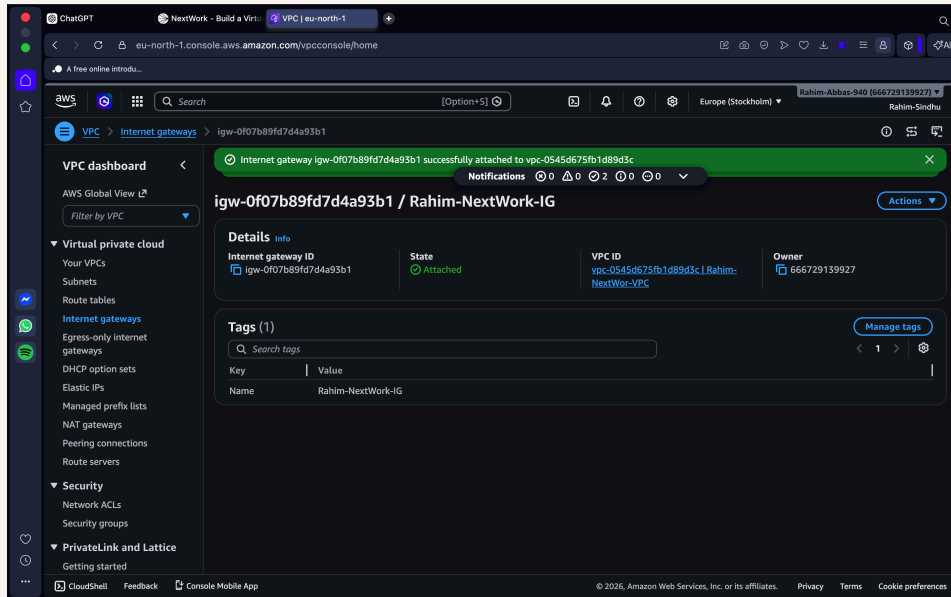
What I did in this step

In this step, I will learn on how to connect my VPC to the internet via a internet gateway to fully implement the theoratical knowledge that I learnt from this project.

Setting up internet gateways

Internet gateways are the bridge that connects the user's VPC to the mainstream internet.

Attaching an Internet Gateway to a VPC enables resources in the VPC to communicate with the internet, provided they are configured with appropriate routing, public IP addresses, and security rules. Without an Internet Gateway, the VPC cannot directly send or receive traffic from the public internet, effectively isolating it from internet access.



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Using the AWS CLI

What I'm doing in this extension

In this project extension, I will launch my VPC in seconds with AWS cloudshell

Exploring CloudShell and CLI

VPC resources could also be created with CloudShell, which an online shell service sitting directly in AWS from where we can run shell commands to achieve our goals. Wherease, AWS CLI is a command line interface that enables us to directly create,manage or delete our AWS services with the help of specific commands.

Debugging my setup

I ran into errors because I was not specifying the CIDR correctly.



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